JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY GURAJADA VIZIANAGARAM

SRI SIVANI COLLEGE OF ENGINEERING (AFFILIATED TO JNTU-GV)

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

★ OFFICIAL ACADEMIC SYLLABUS & BLUEPRINT ★

B.Tech CSE R23 Linear Algebra & Calculus (M1) Official Syllabus & Model Paper

Course Code:	R23-BS-1101	**Academic Scheme:**	I B.Tech I Semester (R23 Regulation)
Verification:	Verified Authentic Academic Copy	**System Ref:**	STUDY-BLUEPRINT-VERIFIED-2026

DETAILED CURRICULAR MODULES & TOPICS

Unit I: Matrices & System of Linear Equations

Rank of matrix by Echelon and Normal forms, Consistency of non-homogeneous system AX=B, Gauss elimination, Gauss-Jordan, and Gauss-Seidel iterative methods.

Unit II: Eigenvalues, Eigenvectors & Cayley-Hamilton Theorem

Eigenvalues and eigenvectors of real matrices, Properties, Cayley-Hamilton theorem and inverse/powers of matrices, Diagonalization, Quadratic forms to canonical forms by orthogonal transformation.

Unit III: Differential Calculus & Mean Value Theorems

Rolle's theorem, Lagrange's and Cauchy's mean value theorems, Taylor's and Maclaurin's series expansions, Indeterminate forms (L'Hopital's rule).

Unit IV: Multivariable Calculus (Partial Differentiation)

Partial derivatives, Total derivative, Chain rule, Jacobians and functional dependence, Maxima and minima of functions of two variables, Lagrange multipliers method.

Unit V: Multiple Integrals & Applications

Double and triple integrals in Cartesian and polar coordinates, Change of order of integration, Change of variables, Evaluation of areas and volumes using multiple integrals.

EVALUATION SCHEME & EXAMINATION BLUEPRINT

Internal Assessment (CIE)	30 Marks
End Examination (SEE)	70 Marks
Credits	3.0 (3-0-0)

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